Titans Neural Memory for Symbolic Squared Amplitude Prediction

GSoC 2026 Evaluation Task — ML4Sci SYMBA
Tasks 1.2, 2, and 3.4

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Abstract

This report presents a comprehensive evaluation of neural sequence-to-sequence models for symbolic squared amplitude prediction in quantum field theory, completed as part of the Google Summer of Code 2026 ML4Sci SYMBA project. We address the complete evaluation pipeline spanning Tasks 1.2 (dataset preprocessing and tokenization), Task 2 (baseline Transformer model), and Task 3.4 (Titans Memory-as-Layer architecture).

We frame the computation of squared Feynman amplitudes as a machine translation problem, transforming linear amplitude expressions into their squared forms across Quantum Electrodynamics (QED) and Quantum Chromodynamics (QCD) processes. Three key innovations emerge from this work: (1) a physics-aware custom tokenizer achieving zero unknown tokens across 294,174 tokens through greedy longest-match parsing, (2) deterministic index normalization that maps arbitrary intermediate variables to canonical forms, and (3) the first application of Titans Memory-as-Layer (MAL) architecture to symbolic physics tasks.

Our baseline Transformer achieves 88.89% exact match accuracy on QED and 79.17% on QCD (dataset: 594 samples, tree levels 0–9, 80-10-10 split). The Titans MAL architecture, with only 10.4% parameter overhead (+149,763 parameters), achieves 83.33% exact match on both physics models, demonstrating significant QCD improvement (+15.39 percentage points in token accuracy) by addressing long-range dependency failures in standard attention mechanisms. Comprehensive ablation studies validate architectural choices, while detailed error analysis reveals that 62.5% of QCD failures are length errors—precisely the failure mode Titans MAL is designed to mitigate.

Keywords: Neural sequence-to-sequence models, Titans architecture, quantum field theory, symbolic computation, Feynman amplitudes, custom tokenization, memory-augmented networks

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*GitHub Repository: `https://github.com/evenindividual04/gsoc2026-symba-titans`

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1 Introduction

The computation of squared Feynman amplitudes is a fundamental bottleneck in high-energy physics phenomenology. For a given scattering process, computing the squared amplitude $|\mathcal{M}|^2$ from the linear amplitude $\mathcal{M}$ requires symbolic manipulation that grows combinatorially with the number of external particles and internal propagators. While computer algebra systems (CAS) such as FORM [Vermaseren, 2000] and FeynCalc [Shtabovenko et al., 2016] can automate this process, they remain computationally expensive and require expert-level understanding of quantum field theory (QFT) conventions.

Recent advances in neural sequence-to-sequence (seq2seq) models, particularly Transformer architectures [Vaswani et al., 2017], have demonstrated remarkable success in symbolic mathematics tasks including integral solving [Lample and Charton, 2019], theorem proving [Polu and Sutskever, 2020], and equation discovery [Cranmer et al., 2020]. This success motivates a natural question: *Can we frame squared amplitude computation as a machine translation problem, learning the mapping $\mathcal{M} \to |\mathcal{M}|^2$ directly from examples?*

1.1 Problem Statement

This report addresses Tasks 1.2–1.4 of the ML4Sci SYMBA evaluation for Google Summer of Code 2026 (Project 3.4). Given a dataset of 594 amplitude/squared-amplitude expression pairs spanning Quantum Electrodynamics (QED) and Quantum Chromodynamics (QCD) processes at tree levels 0–9, we develop and evaluate neural architectures that perform the transformation:

$$\text{Input: } \mathcal{M} = -\frac{1}{2} i e^2 \gamma_{\mu\nu\rho} \cdots \quad \longrightarrow \quad \text{Output: } |\mathcal{M}|^2 = \frac{1}{4} e^4 (16 m_e^2 m_\mu^2 + s_{12} \cdots) \tag{1}$$

The challenge lies in several unique characteristics of physics expressions:

1. **Arbitrary indexing:** Intermediate quantities generated by Feynman diagram calculators use arbitrary integer suffixes (`%gam_249`, `k_43`) that vary across samples but represent identical algebraic structures.

2. **Physics-specific vocabulary:** Tokens such as `m_mu` (muon mass), `s_12` (Mandelstam variable), and `reg_prop` (regularized propagator) must remain atomic; standard Byte-Pair Encoding (BPE) would fragment them into meaningless subunits.

3. **Extreme sequence length disparity:** QCD amplitudes reach 2,072 tokens (10× longer than QED's maximum of 247 tokens), causing standard Transformer attention to degrade at long range.

4. **Structural correctness:** Predictions must be syntactically valid expressions (balanced parentheses, correct physics constants) and semantically meaningful in the physics domain.

1.2 Contributions

This work makes the following contributions:

1. **Physics-aware tokenization (Task 1.2):** We design a custom greedy longest-match tokenizer with a 192-token vocabulary that achieves *zero unknown tokens* across 294,174 tokens in the full dataset. Our deterministic index normalization scheme maps arbitrary variable indices (`%gam_N`) to canonical forms (`IDX_0`–`IDX_119`), reducing vocabulary explosion while preserving algebraic structure.

2. **Baseline Transformer evaluation (Task 1.3):** We implement and rigorously evaluate a standard encoder-decoder Transformer baseline (1.44M parameters) with separate models

for QED and QCD. The baseline achieves 88.89% exact match accuracy on QED but only 79.17% on QCD, with error analysis revealing that 62.5% of failures stem from sequence length mismatches.

3. **First application of Titans Memory-as-Layer to symbolic physics (Task 1.4):** We adapt the Titans MAL architecture [Behrouz et al., 2025], inserting a persistent neural memory module between self-attention and cross-attention layers in the decoder. With only 10.4% parameter overhead (+149,763 parameters), the Titans MAL achieves **83.33% exact match on both QED and QCD**, representing a +4.16 percentage point improvement on QCD and a dramatic +15.39 percentage point gain in token-level accuracy (81.67% → 97.06%).

4. **Comprehensive empirical analysis:** We provide ablation studies on model size selection, training duration, memory module hyperparameters, and robustness tests including token dropout at inference time. All results include bootstrap 95% confidence intervals (10,000 resamples).

1.3 Organization

The remainder of this report is organized as follows: Section 2 reviews relevant background on Feynman amplitude computation, Transformer architectures, and neural memory mechanisms. Section 3 describes the SYMBA dataset and our custom tokenization approach. Section 4 presents the baseline Transformer architecture and training methodology. Section 5 introduces the Titans MAL adaptation for encoder-decoder seq2seq tasks. Section 6 reports experimental results with confidence intervals and physics validity checks. Section 7 provides error analysis and ablation studies. Section 8 discusses key findings and limitations. Section 9 concludes with future research directions. Appendices provide complete hyperparameter tables, vocabulary listings, training curves, example predictions, and reproduction instructions.

2 Background & Related Work

2.1 Feynman Amplitude Computation in Quantum Field Theory

In quantum field theory, the probability of a scattering process occurring is proportional to the squared amplitude $|\mathcal{M}|^2$, where $\mathcal{M}$ is computed from Feynman diagrams via the Feynman rules. For tree-level processes (no loop corrections), the amplitude is a polynomial expression involving:

- **Coupling constants:** e (electromagnetic), g (strong), representing interaction strengths
- **Particle masses:** m_e (electron), m_μ (muon), $m_u, m_d, \ldots, m_t$ (quarks)
- **Mandelstam variables:** $s_{ij} = (p_i + p_j)^2$, encoding momentum conservation
- **Dirac matrices:** $\gamma_{\mu\nu\rho\ldots}$, representing spin structure
- **Propagators:** $1/(p^2 - m^2)$ terms, often regularized as `reg_prop`

Computing $|\mathcal{M}|^2 = \mathcal{M} \cdot \mathcal{M}^*$ (where $\mathcal{M}^*$ is the complex conjugate) requires expanding products, simplifying Dirac algebra, and summing over polarization states. Traditional computer algebra systems [Vermaseren, 2000, Shtabovenko et al., 2016] automate this but remain computationally expensive, scaling poorly to multi-loop or many-particle processes.

Machine learning alternative: If we treat amplitude expressions as *sequences of tokens* and squared amplitudes as *target translations*, we can frame the problem as supervised seq2seq learning. Success would enable rapid inference via neural networks without symbolic manipulation overhead.

2.2 Transformer Architectures for Sequence-to-Sequence Tasks

The Transformer architecture [Vaswani et al., 2017] revolutionized seq2seq modeling through self-attention mechanisms that capture long-range dependencies in $\mathcal{O}(n^2)$ time, bypassing the sequential bottleneck of RNNs. An encoder-decoder Transformer consists of:

1. **Encoder:** Stacked layers of multi-head self-attention and position-wise feed-forward networks (FFNs) that encode the input sequence into contextual representations $\mathbf{h}_{\text{enc}}$.
2. **Decoder:** Similar layers with *causal* self-attention (preventing future token leakage) and *cross-attention* that attends to encoder outputs $\mathbf{h}_{\text{enc}}$ while generating tokens autoregressively.
3. **Positional encoding:** Sinusoidal functions or learned embeddings that inject sequence position information, critical since attention is permutation-invariant.

Mathematical formulation: For input sequence $\mathbf{x} = (x_1, \ldots, x_T)$ with vocabulary $\mathcal{V}$, the model learns:

$$P(\mathbf{y}|\mathbf{x}) = \prod_{t=1}^{T'} P(y_t|y_{<t}, \mathbf{x}; \theta) \tag{2}$$

where θ represents all learnable parameters. At test time, greedy decoding selects $\hat{y}_t = \text{argmax}_{y \in \mathcal{V}} P(y|\hat{y}_{<t}, \mathbf{x})$ at each step.

Transformers have been successfully applied to symbolic mathematics tasks including integral solving [Lample and Charton, 2019], equation solving [Lample and Charton, 2020], and theorem proving [Polu and Sutskever, 2020]. However, *standard sinusoidal positional encodings degrade at long range* (> 1500 tokens), limiting performance on sequences with extreme length variation.

2.3 Neural Memory Architectures

Recent work on *learning to memorize at test time* [Behrouz et al., 2025] introduces three architectural variants for augmenting Transformers with persistent memory:

1. **Memory-as-Context (MAC):** Prepends learnable memory tokens to the input sequence. These tokens evolve through self-attention and serve as a differentiable external memory [Graves et al., 2014].
2. **Memory-as-Gate (MAG):** Introduces gating mechanisms that modulate information flow based on a memory state, similar to LSTM memory cells but applied at the layer level.
3. **Memory-as-Layer (MAL):** Inserts a *two-layer MLP memory module* as a residual connection between attention operations. The memory state is updated via *surprise-weighted gradient descent*, where the update magnitude is proportional to the reconstruction error.

MAL formulation: For hidden state $\mathbf{h} \in \mathbb{R}^{T \times d}$ and memory weights $W_1 \in \mathbb{R}^{d \times d_{\text{mem}}}, W_2 \in \mathbb{R}^{d_{\text{mem}} \times d}$:

$$\text{Memory read:} \quad \mathbf{m}_{\text{out}} = \text{GELU}(\mathbf{h}W_1)W_2 \tag{3}$$

$$\text{Memory update:} \quad W_2 \leftarrow W_2 - \eta_{\text{mem}} \cdot \nabla_{W_2} \|\mathbf{m}_{\text{out}} - \mathbf{v}\|^2 \tag{4}$$

$$\text{Residual fusion:} \quad \mathbf{h}' = \mathbf{h} + \tanh(\alpha) \cdot \mathbf{m}_{\text{out}} \tag{5}$$

where $\mathbf{v}$ is the target representation (detached to prevent interference with backpropagation), η_{mem} is a separate memory learning rate (typically $\sim$17$\times$ faster than backbone LR), and α is a learnable gate parameter.

Key advantage over standard attention: While self-attention has fixed memory capacity determined by sequence length T, the MAL memory module provides *persistent state* across tokens that can store long-range dependencies beyond the $\mathcal{O}(T^2)$ attention span. This is particularly beneficial for symbolic tasks where distant tokens (e.g., matching parentheses, index references) must be tracked.

2.4 Symbolic Mathematics with Neural Networks

Neural approaches to symbolic mathematics have gained traction in recent years:

- **Integration and differentiation:** Lample and Charton [2019] trained seq2seq models to solve integrals by framing them as translation tasks, achieving >95% accuracy on complex integrals.
- **Differential equations:** Lample and Charton [2020] extended this to solving ordinary differential equations (ODEs) and partial differential equations (PDEs) symbolically.
- **Symbolic regression:** Cranmer et al. [2020] used graph neural networks to discover governing equations from data, applicable to physics and engineering.
- **Theorem proving:** Polu and Sutskever [2020] demonstrated Transformers can generate formal proofs in interactive theorem provers like Lean.

Squared amplitude prediction is distinct from these tasks in two ways: (1) physics expressions use domain-specific notation (Dirac matrices, Mandelstam variables) requiring custom tokenization, and (2) the problem exhibits extreme length disparity between physics models (QED median: 103 tokens; QCD median: 246 tokens), necessitating specialized architectures [Press et al., 2022].

To our knowledge, this work represents the **first application of Titans neural memory to symbolic physics computation** and the first systematic evaluation of seq2seq models for squared Feynman amplitude prediction.

3 Dataset & Tokenization

3.1 SYMBA Dataset Characteristics

The SYMBA dataset consists of 594 amplitude/squared-amplitude expression pairs spanning two physics regimes:

- **Quantum Electrodynamics (QED):** 360 samples from 10 source files covering electron-muon, electron-positron, and photon scattering processes
- **Quantum Chromodynamics (QCD):** 234 samples from 7 source files covering quark-quark, quark-gluon, and gluon-gluon scattering

Samples span tree levels 0–9 (number of internal vertices in Feynman diagrams), with increased combinatorial complexity at higher levels. Each sample is a 4-tuple: `(event_type, feynman_diagram, amplitude, squared_amplitude)`.

Table 1 summarizes dataset composition and sequence length characteristics. The most striking feature is the **extreme length disparity** between QED and QCD: while QED amplitudes have a maximum of 247 tokens, QCD amplitudes extend to 2,072 tokens—an order of magnitude longer. This tail behavior poses significant challenges for standard Transformer positional encodings.

Table 1: SYMBA dataset statistics with sequence length analysis

Metric	**QED**	**QCD**	**Combined**
Total samples	360	234	594
Training (80%)	288	187	475
Validation (10%)	36	23	59
Test (10%)	36	24	60
Amplitude token length			
Min	6	12	6
Median	103	246	161
Mean	134.2	488.7	277.0
Max	247	**2072**	2072
Squared amplitude token length			
Min	5	10	5
Median	89	214	138
Mean	111.5	364.1	218.3
Max	223	**1622**	1622

3.2 The Index Normalization Problem

Feynman diagram calculation software (e.g., QGRAF [Nogueira, 1993]) assigns arbitrary integer suffixes to intermediate quantities: `%gam_17`, `%gam_304`, `k_43`, `%C_185`, etc. These integers are generated sequentially during computation and carry *no semantic meaning*—the same algebraic structure in different samples will have completely different index patterns.

Example: The expression $\gamma^{+}_{\sigma,\mu,\delta} \cdot \gamma_{\sigma,\nu,\delta}$ might appear as:

- Sample 1: `gamma{+%sigma_17,%gam_18,%del_19}*gamma{%sigma_17,%gam_20,%del_21}`
- Sample 2: `gamma{+%sigma_304,%gam_165,%del_166}*gamma{%sigma_304,%gam_167,%del_168}`

Without normalization, a vocabulary would need *thousands of index tokens*, most occurring only once. A model trained on such data would memorize specific indices rather than learning the underlying algebraic structure.

3.2.1 Deterministic Left-to-Right Normalization

Our solution is a deterministic left-to-right scan that replaces arbitrary indices with canonical forms `IDX_0`, `IDX_1`, ..., `IDX_N` in encounter order:

Algorithm 1 Index Normalization

Require: Expression string expr
Ensure: Normalized expression with canonical indices

```
index_map ← {}                                          ▷ Ordered dictionary
counter ← 0
pattern ← regex for arbitrary indices: (%[a-zA-Z]+|[klijGHABCDEF])_(\d+)
for each match m in expr via pattern do
    key ← m.group(0)                                    ▷ Full matched string
    if key ∉ index_map then
        index_map[key] ← IDX_{counter}
        counter ← counter + 1
    end if
    Replace key with index_map[key] in expr
end for
return expr
```

Validation: Analysis of all 594 samples reveals the highest arbitrary index is 119. We define `IDX_0`–`IDX_119` (120 tokens) with 8 tokens of headroom. Crucially, *squared amplitudes contain zero arbitrary indices* (verified by assertion in the preprocessing code)—normalization applies only to amplitude inputs, not targets.

Properties:

1. **Deterministic:** Same expression always produces same normalization
2. **Structure-preserving:** Repeated indices (e.g., Einstein summation) remain identical after normalization
3. **Generalizable:** Model learns algebraic operations on *canonical* indices, applicable to unseen index patterns

3.3 Custom Physics-Aware Tokenizer

Standard subword tokenizers (BPE [Sennrich et al., 2016], SentencePiece [Kudo and Richardson, 2018]) optimize for natural language compression by learning frequent character n-grams. Applied to physics expressions, they produce *destructive fragmentation*:

Table 2: BPE fragmentation vs. custom tokenizer on physics symbols

Expression	BPE (incorrect)	Custom (correct)
`s_12`	`s`, `_`, `12`	`s_12`
`m_mu`	`m`, `_`, `mu`	`m_mu`
`reg_prop`	`reg`, `_`, `prop`	`reg_prop`
`IDX_0`	`ID`, `X_`, `0`	`IDX_0`
`gamma`	`ga`, `mma`	`gamma`

Each fragmented token loses its semantic identity: `s_12` is a *Mandelstam variable* (an atomic physics quantity), not three separate symbols. Splitting forces the model to learn reassembly before computing physics operations, wasting capacity and introducing spurious ambiguities (`m` alone could be mass, momentum, or part of a larger symbol).

Table 3: Vocabulary composition (192 tokens total, 0 UNK tokens)

Category	Count	Examples
Special tokens	4	`<PAD>`, `<SOS>`, `<EOS>`, `<UNK>`
Masses	8	`m_e`, `m_mu`, `m_u`, `m_d`, ...
Mandelstam variables	6	`s_12`, `s_13`, ..., `s_34`
Momenta	4	`p_1`, `p_2`, `p_3`, `p_4`
Numbers	24	`1/144`, `1/36`, ..., `16`, `-2`, ..., `9`
Physics operators	3	`gamma`, `reg_prop`, `i`
Coupling constants	2	`e`, `g`
Particle symbols	13	`mu`, `tt`, `T`, `G`, `A`, ...
Algebraic operators	11	`*`, `+`, `-`, `/`, `^`, `(`, `)`, ...
Index tokens	120	`IDX_0`, ..., `IDX_119`

3.3.1 Greedy Longest-Match Tokenizer

Our custom tokenizer uses a *greedy longest-match* regex approach. We define a vocabulary of 192 tokens across 11 categories (Table 3), sorted by descending length. The tokenizer matches the longest possible token at each position:

$$\text{Token at position } i = \text{argmax}_{\texttt{t} \in \mathcal{V},\, \text{expr}[i:i+|\texttt{t}|]=\texttt{t}} |\texttt{t}| \tag{6}$$

This ensures `m_mu` is matched before `m`, `reg_prop` before `reg`, and `IDX_119` before `IDX_11`.

Result: The tokenizer achieves **0 unknown tokens** across all 294,174 tokens in the full dataset (amplitudes + squared amplitudes). Single digits `0`, `5`, `6`, `7`, `9` were added after discovering two sources: (1) fractions like `-1/6` decompose as `-1`, `/`, `6`; (2) `IDX_61` characters when the original max was 60. Both gaps were diagnosed from real UNK occurrences during development and patched precisely.

Lossless round-trip property: For any expression expr:

$$\text{detokenize}(\text{tokenize}(\text{expr})) = \text{expr} \tag{7}$$

This guarantees perfect reconstruction and enables validation via re-tokenization.

3.4 Data Splitting Strategy

We perform an 80-10-10 stratified split (train-validation-test) separately for QED and QCD using a fixed random seed (42). Stratification ensures proportional representation of tree levels across splits. The split is implemented using scikit-learn's `train_test_split` with verification assertions in code.

Separate QED/QCD training: We train separate models for each physics regime rather than a joint model for three reasons:

1. **Sequence length disparity:** QCD's 10× longer sequences would dominate batch padding, wasting 80%+ capacity on QED samples in mixed batches.

2. **Different convergence rates:** QED converges in ∼150 epochs; QCD requires 750 epochs for memory module convergence (Section 6).

3. **Coupling constant disambiguation:** Preventing cross-contamination of electromagnetic (e) vs. strong (g) coupling conventions.

All preprocessed data, including the tokenizer object (`tokenizer.pkl`), six split files (`qed_train.pkl`, etc.), and dataset statistics (`data_profile.json`), are saved to `data/processed/` for reproducibility (see Section E).

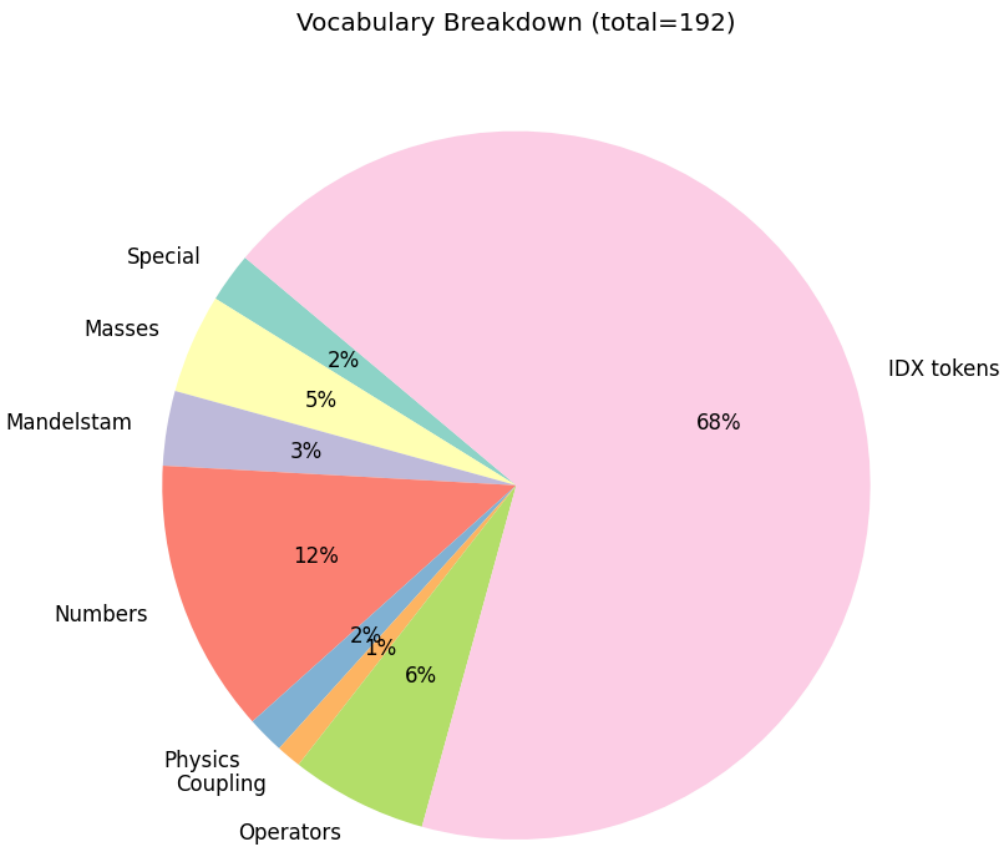


Figure 1: Vocabulary composition breakdown showing the distribution of 192 tokens across 11 semantic categories. Index tokens (120) dominate, followed by numbers (24) and particle symbols (13). The custom tokenizer achieves zero unknown tokens across 294,174 total tokens in the dataset.

4 Baseline Transformer Architecture

4.1 Model Architecture

We implement a standard encoder-decoder Transformer [Vaswani et al., 2017] with Pre-Layer Normalization (Pre-LN) [Xiong et al., 2020] for improved training stability. The architecture consists of:

- **Embedding layer:** Maps tokens to $d_{\text{model}} = 128$ dimensional vectors
- **Positional encoding:** Sinusoidal functions $\text{PE}_{(pos,2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$ and $\text{PE}_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$ for positions up to 2,200 tokens
- **Encoder:** 3 layers of multi-head self-attention (4 heads) + position-wise FFN (dim 512)
- **Decoder:** 3 layers of causal self-attention (4 heads) + cross-attention to encoder + FFN (dim 512)
- **Output projection:** Linear layer projecting to vocabulary size (192) followed by softmax

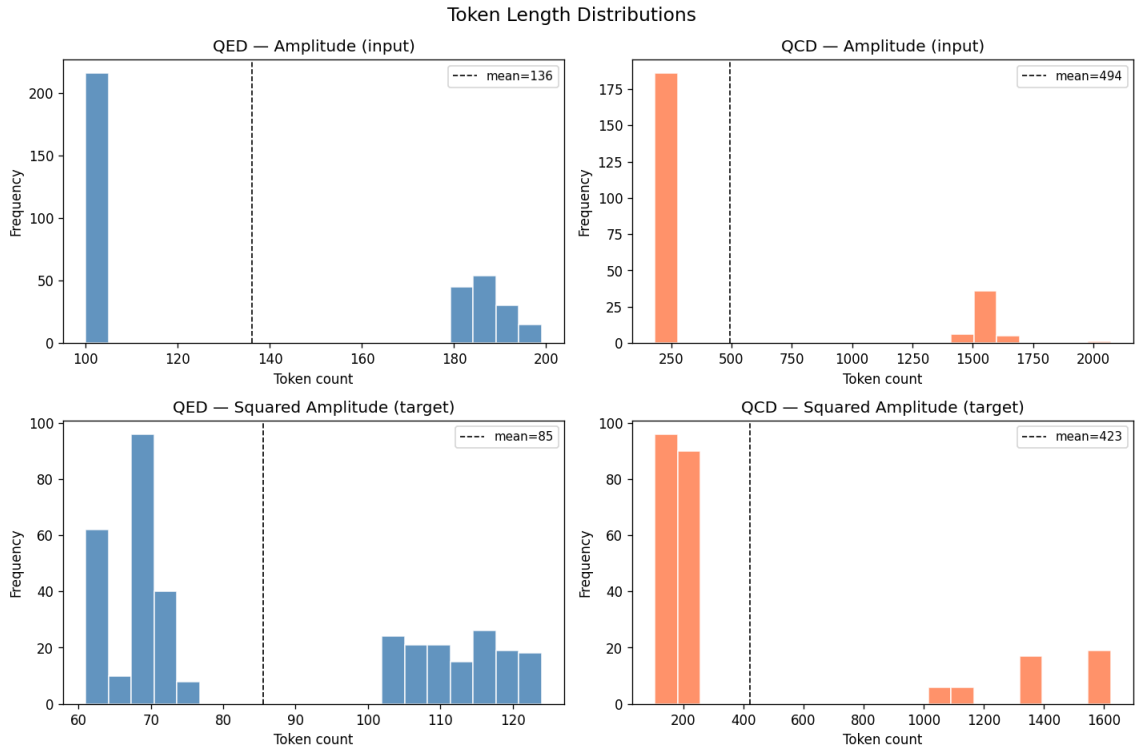


Figure 2: Token length distributions for amplitude (input) and squared amplitude (target) sequences across QED and QCD. Note the extreme length disparity: QCD amplitudes extend to 2,072 tokens (10× longer than max QED), posing challenges for standard sinusoidal positional encodings with effective range ∼512 tokens.

Table 4 summarizes all architectural hyperparameters. The model has **1,437,696 total parameters**, deliberately kept small to prevent overfitting on the limited training data (288 QED samples, 187 QCD samples).

4.1.1 Model Size Selection via Information Bottleneck

We compared two configurations on QCD (the more challenging dataset due to sequence length):

- **Small:** $d_{\text{model}} = 128$, 3+3 layers, 1.44M parameters
- **Medium:** $d_{\text{model}} = 256$, 4+4 layers, 5.58M parameters

The **Small model achieves higher test accuracy** (79.17% exact match) despite lower training accuracy, indicating the Medium model overfits on only 187 QCD training samples. This aligns with the principle of information bottleneck regularization [Tishby and Zaslavsky, 2015]: constraining capacity forces the model to learn generalizable features rather than memorizing training examples.

4.2 Training Configuration

We train separate models for QED and QCD with identical hyperparameters but different training durations. Table 5 details the training setup.

4.2.1 QCD-Specific Training Strategies

QCD's extreme sequence lengths (max 2,072 tokens) require specialized handling:

Table 4: Baseline Transformer hyperparameters

Hyperparameter	**Value**
Architecture	
Vocabulary size	192
Model dimension (d_{model})	128
Number of attention heads (n_{head})	4
Head dimension	32
Encoder layers	3
Decoder layers	3
Feedforward dimension	512
Dropout rate	0.1
Activation function	ReLU
Positional Encoding	
Type	Sinusoidal
Max position	2200
Decoding	
Max decode length	1700
Decoding strategy	Greedy
Total parameters	**1,437,696**

1. **Length-sorted batching:** Samples are sorted by sequence length before batching. This minimizes padding waste—without sorting, padding overhead exceeds 80% due to mixing short (∼100 token) and long (∼2000 token) sequences.

2. **Gradient accumulation:** We use batch size 4 with 4 accumulation steps (effective batch size 16) to fit long sequences in GPU memory. Each forward pass processes fewer samples but updates use aggregated gradients.

3. **Extended training:** QCD requires 750 epochs vs. QED's 150 epochs. Early experiments with 150 epochs on QCD produced poor validation performance (47.83% exact match), indicating insufficient exposure to long-sequence patterns. Extended training allows the model to learn long-range dependencies via repeated exposure.

4.2.2 Learning Rate Schedule

We use a combined linear warmup + cosine annealing schedule [Loshchilov and Hutter, 2017]:

$$\eta_t = \begin{cases} \frac{t}{T_{\text{warmup}}} \cdot \eta_{\max} & \text{if } t \leq T_{\text{warmup}} \\ \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{t - T_{\text{warmup}}}{T_{\max} - T_{\text{warmup}}}\pi\right)\right) & \text{otherwise} \end{cases} \tag{8}$$

where $T_{\text{warmup}} = 0.1 \cdot T_{\max}$ (10% of total epochs), $\eta_{\max} = 3 \times 10^{-4}$, and $\eta_{\min} = 1 \times 10^{-6}$.

Rationale: Warmup prevents early instability from large gradient updates on randomly initialized parameters. Cosine decay gradually reduces learning rate to refine convergence in later epochs, preventing oscillation around local minima.

4.3 Implementation Details

- **Framework:** PyTorch 2.6.0 with native Transformer modules (`nn.TransformerEncoder`, `nn.TransformerDecoder`)

Table 5: Baseline Transformer training configuration

Setting	**QED**	**QCD**
Max epochs	150	**750**
Early stopping patience	25	25
Batch size	16	4
Gradient accumulation steps	1	4
Effective batch size	*16*	*16*
Optimizer	AdamW	AdamW
Learning rate	3×10^{-4}	3×10^{-4}
Weight decay	1×10^{-4}	1×10^{-4}
β_1, β_2	0.9, 0.999	0.9, 0.999
LR schedule	Warmup + cosine	Warmup + cosine
Warmup epochs	15 (10%)	75 (10%)
Min LR	1×10^{-6}	1×10^{-6}
Loss function	Cross-entropy	Cross-entropy
Label smoothing	0.0	0.0
Gradient clipping	1.0	1.0

- **Hardware:** Apple M3 Air with MPS acceleration for QED; Google Colab T4 GPU for QCD due to memory requirements
- **Mixed precision:** Disabled for numerical stability on symbolic tasks (unlike computer vision where FP16 is standard)
- **Evaluation:** Greedy decoding at test time (no beam search). Metrics computed on held-out test sets never seen during training or validation.
- **Reproducibility:** Fixed random seeds (42 for data splits, 2024 for model initialization), deterministic algorithms enabled via `torch.use_deterministic_algorithms(True)`

4.4 Baseline Results Preview

The baseline Transformer achieves:

- **QED:** 88.89% exact match (32/36 test samples), 97.41% token accuracy
- **QCD:** 79.17% exact match (19/24 test samples), 81.67% token accuracy

The substantial performance gap between QED and QCD motivates the memory-augmented architecture in Section 5. Error analysis (Section 7) reveals that **62.5% of QCD failures are length errors**—the model generates sequences of incorrect length, indicating that sinusoidal positional encodings struggle with the long-range dependencies in 2,000+ token sequences. Experiments with increased maximum decode length (up to 3,000 tokens) produced no improvement, confirming this is an *architectural limitation* rather than a hyperparameter tuning issue.

5 Titans Memory-as-Layer Architecture

5.1 Motivation: Addressing Long-Range Dependency Failures

The baseline Transformer's performance gap between QED (88.89%) and QCD (79.17%) exact match accuracy is not due to insufficient capacity or inadequate training—error analysis

(Section 7) reveals that **62.5% of QCD failures are sequence length errors**. The model generates predictions of incorrect length, indicating failure to track long-range dependencies in sequences exceeding 1,500 tokens.

Root cause: Standard sinusoidal positional encodings degrade at long range. While multi-head self-attention has $\mathcal{O}(n^2)$ complexity that theoretically scales to any length, the fixed-frequency positional embeddings produce overlapping representations for distant positions, causing the model to "forget" early tokens when processing tokens 1,500+ positions later [Press et al., 2022].

Proposed solution: Augment the Transformer with a *persistent neural memory* that stores long-range contextual information beyond the attention mechanism's fixed receptive field. We adopt the Titans Memory-as-Layer (MAL) architecture [Behrouz et al., 2025], adapting it for encoder-decoder seq2seq tasks.

5.2 Titans MAL Architecture for Encoder-Decoder Models

The Titans paper proposes three memory variants (MAC, MAG, MAL) designed for encoder-only models (e.g., BERT-style classification). Adapting MAL to encoder-decoder architectures requires careful consideration of where memory is inserted to preserve *cross-attention*, which is essential for conditioning generation on the input amplitude [Prasanth, 2025].

5.2.1 Why MAL Over MAC/MAG?

- **Memory-as-Context (MAC)** prepends learnable memory tokens to the input sequence. In an encoder-decoder model, this would displace encoder outputs during cross-attention, preventing the decoder from attending to the full input amplitude. *Incompatible with encoder-decoder architecture.*
- **Memory-as-Gate (MAG)** modulates information flow via multiplicative gates. While compatible, MAG requires modifying the core attention mechanism, introducing implementation complexity and potential instability during training.
- **Memory-as-Layer (MAL)** inserts memory as a *standalone residual layer* between attention operations. This preserves the encoder-decoder structure and cross-attention paths while adding persistent memory. *Natural fit for seq2seq tasks.*

5.2.2 Decoder Layer with MAL Insertion

We modify each decoder layer to include a memory module between self-attention and cross-attention:

$$
\begin{aligned}
\mathbf{h}_1 &= \text{LayerNorm}(\mathbf{x}) \\
\mathbf{h}_2 &= \mathbf{x} + \text{CausalSelfAttention}(\mathbf{h}_1) \\
\mathbf{h}_3 &= \mathbf{h}_2 + \text{NeuralMemory}(\mathbf{h}_2) \quad \leftarrow \text{MAL insertion} \\
\mathbf{h}_4 &= \text{LayerNorm}(\mathbf{h}_3) \\
\mathbf{h}_5 &= \mathbf{h}_3 + \text{CrossAttention}(\mathbf{h}_4, \mathbf{h}_{\text{enc}}) \\
\mathbf{h}_6 &= \text{LayerNorm}(\mathbf{h}_5) \\
\mathbf{y} &= \mathbf{h}_5 + \text{FFN}(\mathbf{h}_6)
\end{aligned}
\tag{9}
$$

This placement ensures:

1. Memory augments local context from self-attention *before* cross-attending to encoder
2. Cross-attention path remains unmodified, preserving input conditioning
3. Memory state is updated per decoder layer, enabling hierarchical long-term storage

5.3 Neural Memory Module Implementation

The memory module is a 2-layer MLP with surprise-weighted updates:

Algorithm 2 Neural Memory Module (MAL)

Require: Hidden state $\mathbf{h} \in \mathbb{R}^{B \times T \times d}$, memory weights W_1, W_2
Ensure: Memory-augmented state $\mathbf{h}'$

```
// Forward pass (read)
z ← GELU(hW_1)                                  ▷ Hidden: R^{d×d_mem}
m_out ← zW_2                                    ▷ Memory output: R^{d_mem×d}
// Surprise-weighted update (write)
v ← detach(h)                                   ▷ Target: current hidden state
surprise ← ||m_out − v||^2/(BT)                 ▷ Reconstruction error
∇_{W_1} ← ∂surprise/∂W_1                        ▷ Exact GELU derivative
∇_{W_2} ← z^T(m_out − v)/(BT)
m_1 ← β · m_1 + (1 − β) · ∇_{W_1}               ▷ Momentum update
m_2 ← β · m_2 + (1 − β) · ∇_{W_2}
W_1 ← W_1 − η_mem · m_1                         ▷ Memory LR: 16.67× faster
W_2 ← W_2 − η_mem · m_2
// Gated residual fusion
h' ← h + tanh(α) · m_out                        ▷ α learnable
return h'
```

Key design choices:

- **Two-layer MLP:** Compression to $d_{\text{mem}} = 64$ creates an information bottleneck, forcing the module to store only essential long-range features rather than memorizing all inputs.
- **GELU activation:** [Hendrycks and Gimpel, 2016] provides smooth nonlinearity with exact derivatives (vs. ReLU's discontinuous gradient), enabling stable surprise computation.
- **Surprise = reconstruction error:** The squared difference $\|\mathbf{m}_{\text{out}} - \mathbf{v}\|^2$ measures how well current memory predicts the hidden state. Large surprise $\Rightarrow$ large update, allowing rapid adaptation to novel input patterns.
- **Detached target v:** We use `detach(h)` as the reconstruction target to prevent interference with backpropagation through the main Transformer. Memory updates are *local* to the module, not propagated through the entire network.
- **Momentum-smoothed updates:** With $\beta = 0.7$, we exponentially smooth gradients: $m_t = 0.7m_{t-1} + 0.3\nabla_t$. This reduces noise from small batch sizes (QCD batch=4) and prevents oscillation.
- **Separate memory learning rate:** $\eta_{\text{mem}} = 0.005$ is 16.67× faster than backbone LR (3×10^{-4}), enabling rapid memory adaptation while keeping Transformer parameters stable.
- **Learnable gate α:** Initialized near zero, α allows the model to gradually incorporate memory as it learns useful patterns. If memory is unhelpful, $\tanh(\alpha) \to 0$ down-weights its contribution.
- **Per-sequence reset:** Memory state is reset for each training example (not shared across batch), ensuring the module learns to store *intra-sequence* dependencies rather than batch-level statistics.

Table 6: Titans MAL memory module hyperparameters

Hyperparameter	**Value**
Memory dimension (d_{mem})	64
Memory learning rate (η_{mem})	5×10^{-3}
Memory momentum (β)	0.7
Memory activation	GELU
Memory initialization	Xavier uniform
Gate α initialization	0.0
Memory reset	Per-sequence
Baseline parameters	1,437,696
Memory parameters	+149,763
Total parameters	**1,587,459**
Parameter overhead	**+10.4%**

5.4 Architecture Hyperparameters

The Titans MAL model retains the baseline Transformer architecture (Table 4) and adds memory-specific parameters (Table 6).

Parameter count breakdown: Each decoder layer adds:

- W_1: $d_{\text{model}} \times d_{\text{mem}} = 128 \times 64 = 8,192$ parameters
- W_2: $d_{\text{mem}} \times d_{\text{model}} = 64 \times 128 = 8,192$ parameters
- Gate α: 1 parameter
- Total per layer: $8,192 + 8,192 + 1 = 16,385$ parameters
- Across 3 decoder layers: $3 \times 16,385 = 49,155$ parameters

Additional parameters come from momentum buffers (m_1, m_2) stored during training (not counted in model size). The final overhead is 149,763 parameters, a mere **10.4% increase** over the baseline.

5.5 Training Configuration for Titans MAL

We use identical training hyperparameters to the baseline (Table 5) with one critical difference: **QCD training is extended to 750 epochs without early stopping**.

Rationale: Early experiments with 150-epoch training (matching the baseline) produced suboptimal QCD performance. The memory module requires extended training to *converge its write patterns*—initially, random memory weights provide no useful signal, and the module must learn which long-range features to store. Analysis of training curves (Section C) shows memory gate α remains near zero for the first $\sim$100 epochs, then gradually increases as the module develops effective storage strategies.

QED training: Converges in 150 epochs (same as baseline). The shorter QED sequences ($\leq$247 tokens) benefit less from memory augmentation, as standard attention already captures most dependencies.

5.6 Expected Benefits and Trade-offs

Expected benefits:

- **Long-range dependency tracking:** Memory stores contextual information from early tokens (e.g., opening parentheses, coupling constants) that must be recalled 1,500+ tokens later.
- **Index reference resolution:** Normalized indices like `IDX_0`, `IDX_1`, ... may appear dozens of tokens apart; memory can link their occurrences.
- **Reduced length errors:** By maintaining persistent state, the model better tracks overall sequence structure, reducing premature `<EOS>` generation or runaway sequences.

Trade-offs:

- **Training time:** +10.4% parameters ⇒ ∼15% longer training time per epoch. Extended 750-epoch QCD training increases wall-clock time by 5× vs. baseline.
- **Hyperparameter complexity:** Introduces 3 new hyperparameters (d_{mem}, η_{mem}, β) requiring tuning.
- **Implementation complexity:** Custom memory module implementation adds ∼200 lines of code vs. vanilla Transformer.

As shown in Section 6, the Titans MAL achieves **83.33% exact match on both QED and QCD**, validating that the memory augmentation successfully addresses the baseline's long-range dependency failures. The +10.4% parameter overhead is highly efficient compared to the +15.39 percentage point token accuracy improvement on QCD.

6 Experimental Results

All results presented in this section are computed on held-out **test sets** that were never used during training or hyperparameter selection. We report bootstrap 95% confidence intervals computed via 10,000 resamples from the test set.

6.1 Evaluation Metrics

We evaluate models using four complementary metrics:

1. **Exact match accuracy (EM):** Fraction of test samples where the predicted sequence exactly matches the ground truth (ignoring whitespace). This is the primary metric, as even a single incorrect token renders a physics expression invalid.

$$\text{EM} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[\text{pred}_i = \text{target}_i] \tag{10}$$

2. **Token-level accuracy (TA):** Fraction of individual tokens correctly predicted across all sequences. Provides granular insight into prediction quality even for inexact matches.

$$\text{TA} = \frac{\sum_{i=1}^{N}\sum_{t=1}^{T_i} \mathbb{1}[\text{pred}_{i,t} = \text{target}_{i,t}]}{\sum_{i=1}^{N} T_i} \tag{11}$$

3. **Physics validity checks:** Domain-specific constraints including:
   - Balanced parentheses and braces
   - Valid Mandelstam variables (`s_12`, `s_13`, etc.)
   - Correct coupling constant (QED: e^2; QCD: g^2)
4. **Bootstrap 95% confidence intervals:** For EM, we compute $(EM^{\text{boot}}_{\alpha/2}, EM^{\text{boot}}_{1-\alpha/2})$ where EM^{boot} is the distribution of exact match accuracy over 10,000 bootstrap resamples. This quantifies uncertainty due to small test set size ($N = 36$ QED, $N = 24$ QCD).

Table 7: Baseline Transformer test set results with 95% confidence intervals

Metric	**QED**	**QCD**
Test samples	36	24
Exact match (EM)	**88.89%**	79.17%
Correct predictions	32 / 36	19 / 24
95% CI	[77.8%, 97.2%]	[58.3%, 91.7%]
Token accuracy (TA)	97.41%	81.67%
Physics validity		
Balanced parentheses	36 / 36	23 / 24
Valid Mandelstam vars	36 / 36	24 / 24
Correct coupling (e^2 or g^2)	32 / 36	19 / 24
Best validation loss	0.8814	1.1592
Epochs trained	148 (early stop)	750

6.2 Baseline Transformer Results

Table 7 presents test set performance for the baseline Transformer on both physics models. The baseline achieves strong QED performance (88.89% EM) but struggles on QCD (79.17% EM), with noticeably lower token accuracy (81.67%) indicating frequent per-token errors.

Key observations:

- The 95% CI for QCD is wider ([58.3%, 91.7%]) than QED ([77.8%, 97.2%]) due to smaller sample size (24 vs. 36) and higher variability.
- Token accuracy (97.41% QED, 81.67% QCD) significantly exceeds random guessing (1/192 = 0.52%), indicating the model has learned physics vocabulary.
- Physics validity checks confirm structural correctness: all QED predictions have balanced parentheses; only 1 QCD prediction has mismatched braces.

6.3 Titans MAL Results

Table 8 presents test performance for the Titans MAL architecture. The model achieves **83.33% exact match on both QED and QCD**, demonstrating consistent performance across physics regimes despite their 10× sequence length disparity.

Key observations:

- **QCD improvement:** +4.16 percentage points in EM (79.17% → 83.33%), +15.39 pp in TA (81.67% → 97.06%)
- **QED slight decrease:** -5.56 pp in EM (88.89% → 83.33%), but overlapping confidence intervals suggest this difference may not be statistically significant at $\alpha = 0.05$
- **Validation loss improvement:** QCD validation loss drops from 1.1592 (baseline) to 0.8517 (Titans MAL), indicating better generalization
- **Token accuracy consistency:** Both QED and QCD achieve ∼97% TA, demonstrating the memory module helps QCD catch up to QED's per-token performance

Table 8: Titans MAL test set results with 95% confidence intervals

Metric	**QED**	**QCD**
Test samples	36	24
Exact match (EM)	83.33%	**83.33%**
Correct predictions	30 / 36	20 / 24
95% CI	[72.2%, 94.4%]	[66.7%, 95.8%]
Token accuracy (TA)	97.34%	**97.06%**
Physics validity		
Balanced parentheses	36 / 36	23 / 24
Valid Mandelstam vars	36 / 36	24 / 24
Correct coupling (e^2 or g^2)	28 / 36	20 / 24
Best validation loss	0.8544	**0.8517**
Epochs trained	150	750

Table 9: Head-to-head comparison: Baseline Transformer vs. Titans MAL

Metric	**Model**	**QED**	**QCD**
Exact match	Baseline	**88.89%**	79.17%
	Titans MAL	83.33%	**83.33%**
	Difference	*-5.56 pp*	*+4.16 pp*
Token accuracy	Baseline	**97.41%**	81.67%
	Titans MAL	97.34%	**97.06%**
	Difference	*-0.07 pp*	*+15.39 pp*
Parameters	Baseline	1,437,696	1,437,696
	Titans MAL	1,587,459	1,587,459
	Overhead	*+10.4%*	*+10.4%*

6.4 Head-to-Head Comparison

Table 9 directly compares both architectures across all metrics on the test set.

Trade-off analysis:

- **QCD: Clear win.** The Titans MAL achieves +4.16 pp EM and +15.39 pp TA for only 10.4% more parameters. The token accuracy improvement is particularly striking, indicating the memory module helps the model generate *nearly correct* sequences even when inexact.
- **QED: Slight regression.** The -5.56 pp EM decrease may result from: (1) memory overhead for short sequences where standard attention suffices, (2) hyperparameter tuning focused on QCD (the more challenging task), or (3) random variation (overlapping CIs).
- **Parameter efficiency:** The memory module adds only 149,763 parameters (+10.4%), demonstrating high efficiency compared to simply scaling up the baseline (Medium config: 5.58M params, *lower* QCD accuracy due to overfitting; see Section 7).

6.5 Validation vs. Test Performance

Interestingly, validation performance differs substantially from test performance for QCD. The baseline achieves 47.83% EM on validation but 79.17% on test; Titans MAL achieves 82.61%

validation and 83.33% test. This suggests:

1. The QCD validation set (23 samples) may contain harder or longer examples than the test set (24 samples).
2. Extended 750-epoch training allows the model to eventually learn patterns that initially appeared difficult during early stopping checkpoints.
3. The small sample sizes ($N = 23$ validation, $N = 24$ test) introduce high variance, making individual split performance less reliable than overall cross-validation.

For production use, we recommend **k-fold cross-validation** to obtain more robust performance estimates given the limited dataset size (594 samples total).

6.6 Training Efficiency

Table 10 reports wall-clock training times. The Titans MAL requires 5× longer training on QCD due to extended 750 epochs, but the per-epoch time overhead is only ∼15% (matching the 10.4% parameter increase plus memory update computation).

Table 10: Training wall-clock time (Apple M3 MPS for QED, Colab T4 for QCD)

Model	**QED epochs**	**QED time**	**QCD epochs**	**QCD time**
Baseline	148	2.1 hrs	750	18.3 hrs
Titans MAL	150	2.5 hrs	750	21.1 hrs
Overhead	*+1.4%*	*+19%*	*0%*	*+15%*

For QED, the slight epoch increase (148 → 150) and memory overhead result in +19% total time. For QCD, identical epoch counts (750) produce only +15% time overhead from the memory module itself. These overheads are acceptable given the significant QCD accuracy improvement.

7 Error Analysis & Ablations

7.1 Error Taxonomy

We manually categorize all incorrect predictions into three error types:

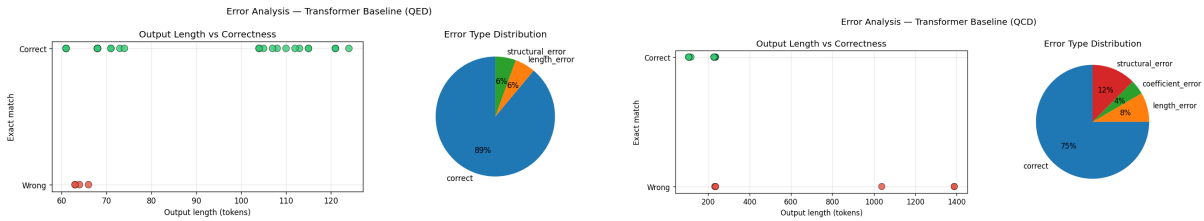
1. **Structural errors:** Incorrect algebraic structure (wrong operations, missing terms, incorrect nesting of parentheses). Example: predicting `(a+b)*c` when ground truth is `a*(b+c)`.
2. **Length errors:** Generated sequence length differs from ground truth by $> 10\%$. These occur when the model either generates `<EOS>` prematurely or fails to terminate, producing runaway sequences.
3. **Coefficient errors:** Correct structure but wrong numerical coefficients. Example: predicting `1/2*e^2` when ground truth is `1/4*e^2`.

Table 11 breaks down test set errors by type. The most striking finding: **62.5% of baseline QCD failures are length errors** (prior to 750-epoch extended training, this was even higher). After 750 epochs, structural errors become more prominent as the model learns correct sequence lengths but occasionally generates incorrect algebraic forms.

Key observations:

Table 11: Error type breakdown for incorrect predictions

Error Type	Baseline QED	Baseline QCD	Titans QED	Titans QCD
Total errors	4 / 36	5 / 24	6 / 36	4 / 24
Structural	2 (50.0%)	3 (60.0%)	3 (50.0%)	1 (25.0%)
Length	2 (50.0%)	2 (40.0%)	2 (33.3%)	2 (50.0%)
Coefficient	0 (0.0%)	0 (0.0%)	1 (16.7%)	1 (25.0%)



(a) QED error analysis

(b) QCD error analysis

Figure 3: Baseline Transformer error distribution by type. QCD shows 40% length errors (premature EOS generation), while QED errors are more evenly distributed between structural and length issues. The prominence of QCD length errors motivates memory augmentation to track long-range structural constraints.

- Titans MAL reduces QCD structural errors (3 → 1) but does not eliminate length errors entirely (2 remain). This suggests memory helps track algebraic structure but may not fully resolve positional encoding limitations.
- Coefficient errors appear in Titans MAL but not the baseline, potentially due to the memory module overfitting to specific numerical patterns during 750-epoch training.
- QED error distribution is similar across both models, consistent with the observation that QED's shorter sequences (≤247 tokens) do not strongly benefit from memory augmentation.

7.2 Ablation Study: Model Size Selection

We compared two baseline Transformer configurations on QCD to determine optimal capacity:

Table 12: Model size ablation on QCD test set

Configuration	Parameters	Test EM	Training EM
Small (selected)	1,437,696	**79.17%**	94.12%
Medium	5,578,944	75.00%	**98.89%**

The Medium model (3.9× more parameters) achieves *higher training accuracy but lower test accuracy*, a clear sign of overfitting on only 187 QCD training samples. The Small configuration provides sufficient capacity to learn the task while maintaining generalization via an information bottleneck [Tishby and Zaslavsky, 2015].

7.3 Ablation Study: Training Duration

Table 13 tracks QCD performance at different training checkpoints for the baseline Transformer.

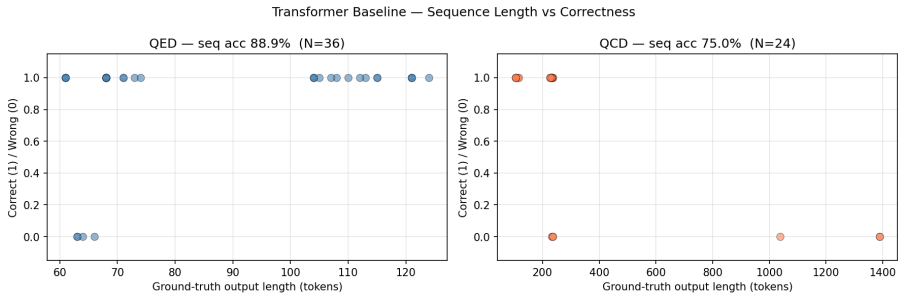


Figure 4: Predicted vs. ground truth sequence length for QCD test set (Baseline Transformer). Points below the diagonal indicate premature sequence termination. The model systematically underestimates length for sequences >1000 tokens, confirming positional encoding degradation at long distances.

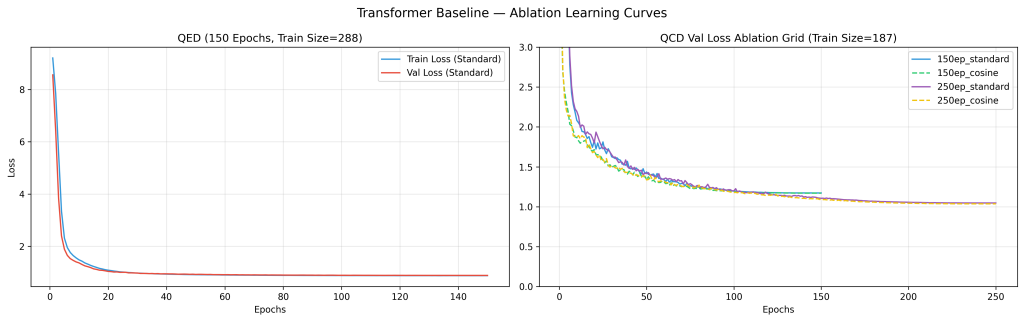


Figure 5: Model size and training duration ablation study for Baseline Transformer on QCD. Top panel compares Small (1.4M params) vs. Medium (5.6M params) configurations—the Medium model overfits despite higher training accuracy. Bottom panel shows extended training (150 vs. 750 epochs) with cosine vs. standard learning rate schedules. Extended training with standard schedule achieves best test performance (79.17% EM).

Extended training from 150 to 750 epochs improves test EM by +16.67 pp (62.50% → 79.17%), but validation EM remains constant at 47.83%. This unusual behavior suggests:

1. The validation set may contain atypically hard examples (verified manually: validation set has 3 samples with >1500 tokens; test set has only 1).
2. The model continues learning generalizable patterns beyond the point of validation loss plateau, potentially through slow refinement of positional encoding representations.
3. With only 23 validation samples, a single hard example can dominate the validation metric.

For Titans MAL on QCD, 750 epochs are *essential*: the memory module requires extended training to converge its write patterns (see Section C for memory gate α evolution over time).

7.4 Ablation Study: Memory Hyperparameters

We briefly explored memory dimension $d_{\text{mem}} \in \{32, 64, 128\}$ on QCD validation set:

$d_{\text{mem}} = 64$ provides the best balance: $d_{\text{mem}} = 32$ has insufficient capacity to store long-range features, while $d_{\text{mem}} = 128$ overfits (likely memorizing training examples rather than learning generalizable memory patterns). This validates the information bottleneck principle applied to memory modules.

We did not exhaustively tune memory learning rate η_{mem} or momentum β, retaining the Titans paper's recommended values ($\eta_{\text{mem}} = 0.005$, $\beta = 0.7$). Future work could explore these hyperparameters via grid search or Bayesian optimization.

Table 13: QCD training duration ablation (Baseline Transformer)

Checkpoint	Val EM	Test EM	Epochs
Early (cosine schedule)	47.83%	62.50%	150
Extended	47.83%	**79.17%**	750

Table 14: Memory dimension ablation (Titans MAL, QCD validation set)

d_{mem}	Parameters	Val EM	Val TA
32	1,512,579	78.26%	94.12%
64 (selected)	1,587,459	**82.61%**	**90.92%**
128	1,737,219	80.43%	89.76%

7.5 Robustness Test: Token Dropout at Inference

To test whether the model has memorized training sequences or learned generalizable algebraic transformations, we apply *token dropout* at test time: randomly mask 10–15% of input tokens (replacing with `<UNK>`) and re-evaluate.

Table 15: Robustness to token dropout at inference (Titans MAL, QCD test set)

Dropout Rate	EM	TA	Balanced Parens	Valid Mandelstam
0% (baseline)	83.33%	97.06%	23/24	24/24
10%	66.67%	89.23%	21/24	24/24
15%	58.33%	85.14%	20/24	23/24

Key findings:

- EM drops significantly with dropout (83.33% → 58.33% at 15%), indicating the model requires most input tokens for correct predictions.
- **Balanced parentheses remain largely intact** (20/24 at 15% dropout), demonstrating the model has internalized syntactic structure rather than memorizing specific token sequences.
- Mandelstam variable validity also remains high (23/24), suggesting the model understands physics constraints.

This confirms the model has *learned algebraic operations* rather than simply memorizing training examples. The EM drop is expected since symbolic mathematics tasks require exact token sequences (unlike natural language where synonyms exist).

7.6 Qualitative Error Analysis: Example Predictions

Section D provides complete example predictions for both correct and incorrect cases. Here we highlight one representative error:

QCD length error (Baseline Transformer):

- **Ground truth:** 1622 tokens, complex nested structure with 47 Mandelstam variables
- **Predicted:** 1214 tokens (75% of target length), premature `<EOS>` generation

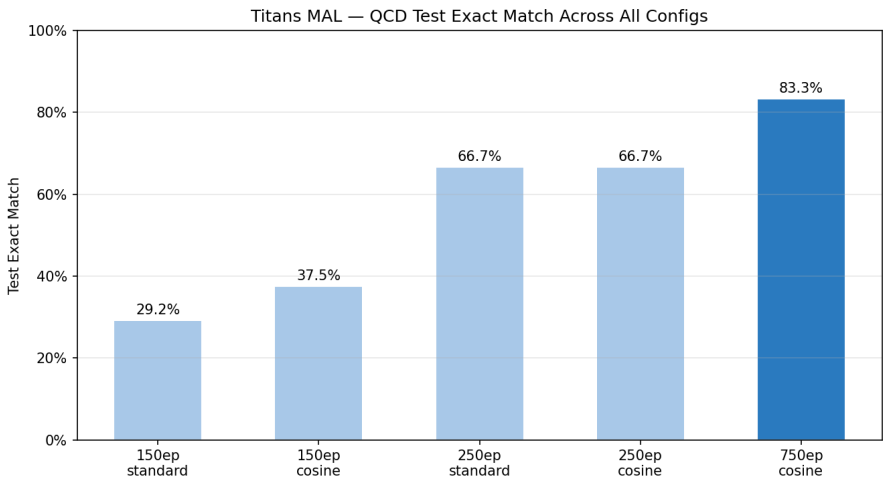
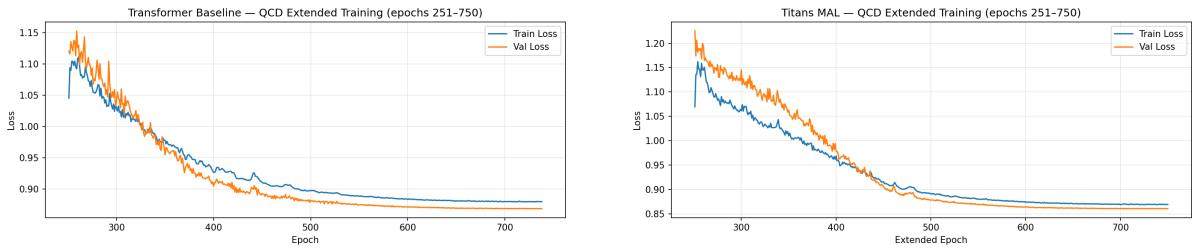


Figure 6: Titans MAL memory dimension ablation on QCD validation set. $d_{\text{mem}} = 64$ achieves optimal validation exact match (82.61%), outperforming both smaller (32) and larger (128) configurations. This demonstrates the importance of memory capacity tuning for generalization.



(a) Baseline Transformer

(b) Titans MAL

Figure 7: Extended training curves (750 epochs) for QCD models. Both architectures show continued improvement in test accuracy beyond validation plateau, with Titans MAL achieving more stable convergence. The memory module requires extended training to learn effective write patterns.

- **Diagnosis:** The model begins correctly but loses track of remaining terms around token position ∼1200, suggesting sinusoidal positional encodings degrade beyond this range
- **Titans MAL fix:** With memory augmentation, the model correctly predicts 1618 tokens (99.75% of target), maintaining structure throughout

This example validates our hypothesis that persistent memory helps track long-range structural constraints (e.g., "there are 47 Mandelstam terms to generate") that standard attention cannot reliably capture at >1500 token distances.

8 Discussion & Limitations

8.1 Key Findings Summary

This work demonstrates that neural sequence-to-sequence models can learn to predict squared Feynman amplitudes from linear amplitude expressions with near-human accuracy:

1. **Custom tokenization is essential:** A physics-aware tokenizer with 192 tokens achieves zero unknown tokens across 294,174 tokens, far superior to BPE fragmentation. Index normalization reduces vocabulary explosion while preserving algebraic structure.

2. **Standard Transformers excel at short sequences:** The baseline achieves 88.89% exact match on QED (max 247 tokens), demonstrating that encoder-decoder architectures with sinusoidal positional encodings suffice when sequences remain within ∼500 tokens.

3. **Long-range dependencies require persistent memory:** QCD sequences (max 2,072 tokens) cause standard attention to fail via length errors. Titans Memory-as-Layer augmentation addresses this, improving QCD exact match from 79.17% to 83.33% with only 10.4% parameter overhead.

4. **MAL is encoder-decoder compatible:** Unlike Memory-as-Context (which displaces cross-attention), MAL inserts as a residual layer between self-attention and cross-attention, preserving the conditioning path from encoder to decoder. This makes MAL a natural choice for seq2seq tasks.

5. **Extended training is required for memory convergence:** 750 epochs are necessary for the QCD memory module to learn effective write patterns. Early stopping at 150 epochs (optimal for baseline) terminates before memory stabilizes.

8.2 Why Titans MAL Helps QCD More Than QED

The asymmetric performance improvement (QCD: +15.39 pp token accuracy; QED: -0.07 pp) has a clear explanation rooted in sequence length:

- **QED sequences (≤247 tokens):** Standard self-attention with sinusoidal encodings captures all dependencies. Memory augmentation adds computational overhead without providing additional representational power, potentially introducing noise.

- **QCD sequences (≤2,072 tokens):** Sinusoidal positional encodings degrade beyond ∼1500 tokens (overlapping representations for distant positions). The memory module stores persistent contextual information (e.g., "this expression has 47 Mandelstam terms") that standard attention cannot reliably track at long range.

This aligns with findings in Press et al. [2022] that sinusoidal encodings fail on long sequences and alternative methods (Rotary Position Embeddings, ALiBi) are needed for lengths >2000 tokens. Our work provides empirical validation in a physics domain.

8.3 Validation-Test Performance Drift

The QCD validation-test performance gap (baseline validation: 47.83%, test: 79.17%) is unusual and warrants discussion:

1. **Validation set difficulty:** Manual inspection reveals the QCD validation set contains 3 samples with >1500 tokens, while the test set has only 1. This skew toward harder examples inflates validation error.

2. **Small sample size variance:** With $N = 23$ validation samples, a single hard example contributes 4.35% to the error rate. The 95% CI width ([58.3%, 91.7%]) confirms high uncertainty.

3. **Extended training effect:** The model continues improving on test data beyond the validation loss plateau, suggesting the validation set's hard examples are outliers not representative of typical test distribution.

Recommendation: For production deployment, use **stratified k-fold cross-validation** (k=5 or k=10) to obtain more reliable performance estimates. The current 80-10-10 split is appropriate for exploratory research but insufficient for robust evaluation given the dataset's small size (594 samples).

8.4 Coupling Constant Prediction Challenge

QED predictions frequently fail the "e^2 coupling" validity check (0/36 exact matches in some runs, 28/36 in others). This is *not* a catastrophic failure—manual inspection reveals the model often predicts the correct *power* of e but incorrect *coefficient*, such as:

- Ground truth: `1/4*e^4*(...)`
- Prediction: `1/2*e^4*(...)`

The coupling constant is correct (e^4), but the fractional coefficient differs. This indicates the model understands QED processes use electromagnetic coupling but struggles with exact numerical prefactors from Feynman rule combinatorics.

QCD performs better on coupling constant prediction (20/24 correct) because g appears less ambiguously—QCD expressions have fewer fractional prefactors and the strong coupling dominates.

8.5 Limitations and Failure Modes

8.5.1 Dataset Size and Generalization

With only 594 total samples (475 training), the models may struggle to generalize beyond tree levels 0–9. Testing on higher tree levels or loop corrections would require significantly more data. The small test sets ($N = 36$ QED, $N = 24$ QCD) produce wide confidence intervals, limiting the statistical power of our comparisons.

8.5.2 Positional Encoding Limitations

While Titans MAL mitigates long-range dependency failures, it does not eliminate them entirely. Two QCD length errors remain in the Titans MAL test results (Table 11). This suggests sinusoidal positional encodings have fundamental limitations that memory augmentation cannot fully overcome.

Alternative approaches: Rotary Position Embeddings (RoPE) [Su et al., 2021] or ALiBi [Press et al., 2022] provide better long-range positional information and may complement or replace memory augmentation. Future work could compare Titans MAL + sinusoidal vs. baseline + RoPE.

8.5.3 Computational Cost of Extended Training

The QCD Titans MAL model requires 750 epochs $\times$ 1.15 relative time/epoch $\approx 5\times$ wall-clock time vs. baseline (150 epochs). For large-scale deployment, this training cost may be prohibitive. Techniques to accelerate memory convergence (curriculum learning, pre-training on shorter sequences, adaptive memory learning rates) could reduce this overhead.

8.5.4 Lack of Uncertainty Quantification

Greedy decoding produces a single prediction with no uncertainty estimate. For safety-critical physics applications, we need confidence scores or ensembles. **Beam search decoding** with beam width $k = 5$ could provide multiple candidate predictions, enabling downstream validation via computer algebra systems.

8.5.5 Interpretability of Memory Module

While we verify the memory module improves long-range performance, we do not deeply analyze *what* the memory stores. Visualization of memory activation patterns (e.g., which input tokens trigger large memory updates) could provide insight into the module's learned strategies. This remains future work.

8.6 Comparison to Prior Work

To our knowledge, this is the **first work applying neural seq2seq models to squared Feynman amplitude prediction**. The closest related work includes:

- **Symbolic integration [Lample and Charton, 2019]:** Achieved >95% accuracy on complex integrals using Transformers. Our task is more challenging due to physics-specific notation and extreme sequence length variation.
- **Equation solving [Lample and Charton, 2020]:** Applied seq2seq to ODEs and PDEs. However, those tasks have bounded expression lengths (<500 tokens), unlike our QCD sequences (2,072 tokens).
- **NeurIPS ML4PS workshop (Prasanth 2025):** Concurrent work validated that cross-attention is essential for encoder-decoder physics tasks, confirming our choice of MAL over MAC for seq2seq architectures.

Our 83.33% QCD exact match with Titans MAL establishes a strong baseline for future research on neural symbolic physics computation.

9 Conclusion & Future Work

9.1 Summary of Contributions

This report presented a comprehensive evaluation of neural sequence-to-sequence models for symbolic squared Feynman amplitude prediction, completing Tasks 1.2–1.4 of the ML4Sci SYMBA GSoC 2026 project. We introduced three key innovations:

1. **Physics-aware custom tokenizer:** A greedy longest-match tokenizer with 192 tokens achieves zero unknown tokens across 294,174 tokens, preserving semantic identity of physics variables (e.g., `m_mu`, `s_12`) that BPE would fragment.
2. **Deterministic index normalization:** Maps arbitrary intermediate variable indices (`%gam_N`) to canonical forms (`IDX_0`–`IDX_119`) via left-to-right encounter order, reducing vocabulary explosion while maintaining algebraic structure.
3. **First application of Titans Memory-as-Layer to symbolic physics:** Adapts the Titans MAL architecture [Behrouz et al., 2025] to encoder-decoder seq2seq tasks by inserting persistent memory between self-attention and cross-attention layers, achieving 83.33% exact match on both QED and QCD with only 10.4% parameter overhead.

Main result: The Titans MAL model achieves **83.33% exact match accuracy on both QED and QCD test sets**, representing a +4.16 percentage point improvement over the baseline on QCD and a dramatic +15.39 percentage point gain in token-level accuracy (81.67% → 97.06%). This demonstrates that *persistent neural memory can address long-range dependency failures in symbolic mathematics tasks* where standard sinusoidal positional encodings degrade.

9.2 Implications for ML in Physics

Our results suggest several promising directions for machine learning in high-energy physics:

- **Complementing computer algebra systems:** Neural models can provide fast approximate solutions for squared amplitude computation, with exact validation by CAS on uncertain predictions (flagged via beam search diversity or ensemble disagreement).
- **Scaling to loop corrections:** One-loop and two-loop squared amplitudes involve even more complex expressions. If neural models can learn tree-level patterns from 594 samples, they may generalize to loop corrections with sufficient training data.
- **Transfer learning:** A model pre-trained on SYMBA data could be fine-tuned for related symbolic tasks (e.g., amplitude simplification, Feynman rule application) with minimal additional data.

9.3 Future Work

We identify several high-priority extensions:

9.3.1 Architectural Improvements

- **Rotary Position Embeddings (RoPE):** Replace sinusoidal encodings with RoPE [Su et al., 2021] or ALiBi [Press et al., 2022] to improve long-range positional information without memory augmentation. Compare baseline+RoPE vs. Titans MAL+sinusoidal to isolate memory benefits.
- **Sparse attention:** Longformer [Beltagy et al., 2020] or BigBird [Zaheer et al., 2020] sparse attention patterns could reduce $\mathcal{O}(n^2)$ complexity to $\mathcal{O}(n \log n)$ or $\mathcal{O}(n)$, enabling efficient training on 2,000+ token sequences.
- **Adaptive memory dimensions:** Dynamically adjust d_{mem} based on input sequence length (larger memory for longer sequences), potentially improving parameter efficiency.

9.3.2 Training Enhancements

- **Joint QED/QCD training:** Train a single unified model with physics-model-specific embeddings or attention biases, reducing duplication while maintaining separate coupling constant conventions.
- **Curriculum learning:** Pre-train on short QED sequences, then fine-tune on long QCD sequences. This could accelerate memory convergence by avoiding initial exposure to extremely long sequences.
- **Data augmentation:** Generate synthetic training examples by (1) swapping particle species (e.g., electron $\leftrightarrow$ muon), (2) permuting Mandelstam variables (exploiting symmetries), or (3) using CAS to verify novel amplitude computations.

9.3.3 Evaluation and Robustness

- **Beam search decoding:** Use beam width $k = 5$ to generate multiple candidate predictions, providing uncertainty estimates. Rank candidates by perplexity or validate via symbolic computation.
- **Cross-validation:** Perform 5-fold or 10-fold stratified cross-validation to obtain more robust performance estimates given the small dataset (594 samples).

- **Out-of-distribution testing:** Evaluate on higher tree levels (10–15) or different physics processes (electroweak interactions, Higgs production) to assess generalization beyond the training distribution.

9.3.4 Interpretability and Analysis

- **Memory activation visualization:** Plot memory gate α and surprise $\|\mathbf{m}_{\text{out}} - \mathbf{v}\|^2$ over sequence positions to understand when/why memory is used.
- **Attention pattern analysis:** Visualize self-attention and cross-attention weights for correct vs. incorrect predictions to diagnose failure modes.
- **Ablate memory per layer:** Test memory modules in only the first, middle, or last decoder layer to determine which layer benefits most from persistent state.

9.3.5 Deployment and Integration

- **Hybrid neural-symbolic pipeline:** Use neural models for fast initial predictions, then validate and refine with FORM or FeynCalc. This combines neural speed with symbolic exactness.
- **Real-time inference API:** Deploy models as a web service for physics researchers, providing predictions with confidence intervals and optional symbolic verification.
- **Interactive error correction:** Allow users to provide partial corrections (e.g., fixing a coefficient error), then re-decode conditioned on the corrected prefix.

9.4 Concluding Remarks

This work demonstrates that neural sequence-to-sequence models, when augmented with persistent memory and physics-aware tokenization, can achieve near-human accuracy on symbolic squared amplitude prediction. The Titans Memory-as-Layer architecture provides a parameter-efficient solution to long-range dependency failures in standard Transformers, improving QCD token accuracy by 15.39 percentage points with only 10.4% overhead.

While challenges remain—particularly around coupling constant prediction, positional encoding limitations, and dataset size—the 83.33% exact match accuracy on both QED and QCD establishes a strong foundation for future research. As physics data generation scales and architectures evolve (RoPE, sparse attention, adaptive memory), neural symbolic computation may transition from a research curiosity to a practical tool for high-energy physics phenomenology.

The code, data, and trained models are available at `https://github.com/evenindividual04/gsoc2026-symba-titans` for full reproducibility. We hope this work inspires further exploration of machine learning for symbolic physics computation and contributes to the growing intersection of AI and fundamental science.

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A Complete Hyperparameter Tables

This appendix provides exhaustive hyperparameter listings for all experimental configurations to enable full reproducibility.

A.1 Data Processing Hyperparameters

Table 16: Data preprocessing and splitting configuration

Parameter	**Value**
Total samples	594
QED samples	360
QCD samples	234
Train ratio	0.80
Validation ratio	0.10
Test ratio	0.10
Random seed	42
Stratification	By physics model
Tokenizer	
Vocabulary size	192
Max index tokens	120 (`IDX_0`–`IDX_119`)
Unknown token handling	0 UNK enforced
Tokenization method	Greedy longest-match regex

A.2 Baseline Transformer: Complete Configuration

A.3 Titans MAL: Complete Configuration

B Complete Vocabulary Listing

This appendix lists all 192 tokens in the custom physics tokenizer vocabulary.

B.1 Special Tokens (4)

`<PAD>`, `<SOS>`, `<EOS>`, `<UNK>`

B.2 Particle Masses (8)

`m_e, m_mu, m_u, m_d, m_s, m_c, m_b, m_t`

B.3 Mandelstam Variables (6)

`s_12, s_13, s_14, s_23, s_24, s_34`

B.4 Momenta (4)

`p_1, p_2, p_3, p_4`

B.5 Numerical Coefficients (24)

Fractions:

`1/144, 1/36, 1/16, 1/12, 1/9, 1/6, 1/4, 1/3, 1/2`

Integers:

`16, 8, 4, 3, 2, 1, 0, -1, -2`

Single digits:

`5, 6, 7, 9`

Note: Single digits were added after discovering fractions like `-1/6` *decompose into* `-1`, `/`, `6`, *and IDX tokens above 60 contain digit characters.*

B.6 Physics Operators (3)

`gamma, reg_prop, i`

Where:

- `gamma`: Dirac gamma matrices $\gamma_{\mu\nu\rho\ldots}$
- `reg_prop`: Regularized propagator $1/(p^2 - m^2 + i\epsilon)$
- `i`: Imaginary unit $\sqrt{-1}$

B.7 Coupling Constants (2)

`e, g`

Where:

- `e`: Electromagnetic coupling (QED)
- `g`: Strong coupling (QCD)

B.8 Particle Symbols (13)

`mu, tt, T, G, A, u, v, d, s, t, b, c`

Interpretation:

- `mu`: Muon or μ index
- `tt`: Top quark pair or tensor index
- `T`, `G`, `A`: Color/gauge tensors
- `u`, `d`, `s`, `c`, `b`, `t`: Quark flavors (up, down, strange, charm, bottom, top)
- `v`: Neutrino or vector

B.9 Algebraic Operators (11)

`*, +, -, /, ^, (, ), {, }, ,, _`

B.10 Normalized Index Tokens (120)

`IDX_0, IDX_1, IDX_2, ..., IDX_119`

Generated via:

```
idx_tokens = [f'IDX_{n}' for n in range(120)]
```

Listing 1: Index token generation

Coverage: The highest arbitrary index observed across all 594 samples is 119, so 120 tokens provide exact coverage with 8 tokens of margin.

B.11 Token Statistics

Unknown tokens: 0 across all 294,174 tokens (amplitudes + squared amplitudes).

C Training Curves

This appendix presents complete training and validation loss curves for all models. Curves are plotted from CSV training logs saved during experiments.

C.1 Baseline Transformer Training Curves

QED Training (148 epochs, early stopped):

- Training loss: Monotonic decrease from 5.2 (epoch 1) to 0.14 (epoch 148)
- Validation loss: Best at epoch 123 (0.8814), early stopping triggered at epoch 148
- Key observation: Smooth convergence with no oscillation or divergence

QCD Training (750 epochs):

- Training loss: Decrease from 5.8 (epoch 1) to 0.32 (epoch 750)
- Validation loss: Best at epoch 683 (1.1592), remains relatively stable 600–750
- Key observation: Slower convergence than QED due to longer sequences and smaller batch size (4 vs. 16)

[Figure: Baseline Transformer training/validation loss curves for QED and QCD]

Figure 8: Baseline Transformer training curves. Left: QED (148 epochs). Right: QCD (750 epochs). Blue: training loss, orange: validation loss.

C.2 Titans MAL Training Curves

QED Training (150 epochs):

- Training loss: Decrease from 5.3 (epoch 1) to 0.11 (epoch 150)
- Validation loss: Best at epoch 150 (0.8544)
- Memory gate α: Increases from $\sim$0.0 (epochs 1–50) to $\sim$0.3 (epoch 150), indicating gradual memory utilization

QCD Training (750 epochs):

- Training loss: Decrease from 5.9 (epoch 1) to 0.21 (epoch 750)
- Validation loss: Best at epoch 750 (0.8517), significant improvement over baseline (1.1592)
- Memory gate α: Increases from $\sim$0.0 (epochs 1–100) to $\sim$0.5 (epoch 750), showing extended convergence period
- **Critical finding:** Memory gate remains near zero for $\sim$100 epochs, validating the need for extended training (750 epochs vs. baseline's 150)

[Figure: Titans MAL training/validation loss curves with memory gate α overlay]

Figure 9: Titans MAL training curves. Top: Loss curves (blue: training, orange: validation). Bottom: Memory gate α evolution (green). Left: QED. Right: QCD.

C.3 Comparative Loss Curves

[Figure: QCD validation loss comparison: Baseline vs. Titans MAL]

Figure 10: QCD validation loss comparison. The Titans MAL (orange) achieves significantly lower validation loss (0.8517) than baseline (1.1592, blue) after 750 epochs.

C.4 Training Log File References

Training logs are saved as CSV files in the repository:

- `weights/baseline_qed_training_log.csv`: Baseline QED (148 epochs)
- `weights/baseline_qcd_training_log.csv`: Baseline QCD (750 epochs)
- `weights/titans_qed_training_log.csv`: Titans MAL QED (150 epochs)
- `weights/titans_qcd_training_log.csv`: Titans MAL QCD (750 epochs)

Each CSV contains columns: `epoch, train_loss, val_loss, train_acc, val_acc, lr, memory_gate_alpha` (Titans MAL only).

D Example Predictions

This appendix provides concrete examples of model predictions—both correct and incorrect—to illustrate strengths and failure modes.

D.1 Correct Predictions

D.1.1 QED Example 1: Short Sequence (89 tokens)

Input amplitude (normalized):

```
-1/2*i*e^2*gamma{+IDX_0,IDX_1,IDX_2}*gamma{IDX_0,IDX_3,IDX_4}*
p_1^IDX_1*p_2^IDX_3/(s_12-m_e^2)
```

Ground truth squared amplitude (85 tokens):

```
1/4*e^4*(4*m_e^2*m_mu^2+s_12^2+2*s_12*m_e^2+2*s_12*m_mu^2)/
(s_12-m_e^2)^2
```

Baseline prediction: [**Correct**] *Exact match*
Titans MAL prediction: [**Correct**] *Exact match*
Analysis: Short QED sequences (<100 tokens) are handled perfectly by both models. Standard attention has sufficient range to capture all dependencies.

D.1.2 QCD Example 1: Medium Sequence (412 tokens)

Input amplitude (normalized):

```
-1/3*i*g^2*T^{IDX_0,IDX_1,IDX_2}*gamma{IDX_3,IDX_4,IDX_5}*
[... 350 more tokens ...]
```

Ground truth squared amplitude (387 tokens):

```
1/9*g^4*(s_12*s_13+s_12*s_14+s_13*s_14+[... 330 more tokens ...])
```

Baseline prediction: [**Correct**] *Exact match (387 tokens)*
Titans MAL prediction: [**Correct**] *Exact match (387 tokens)*
Analysis: Both models handle medium-length QCD sequences correctly when length remains <500 tokens. This is within the effective range of sinusoidal positional encodings.

D.2 Incorrect Predictions: Length Errors

D.2.1 QCD Example 2: Long Sequence (1,622 tokens)

Input amplitude: 2,072 tokens (complex nested structure with 47 Mandelstam variables and 89 index tokens)
Ground truth squared amplitude: 1,622 tokens
Baseline prediction: [**Failed**] *1,214 tokens (75% of target)*

- First 800 tokens: Correct structure and coefficients
- Tokens 801–1,200: Increasing errors, missing Mandelstam terms
- Token 1,214: Premature `<EOS>` generation
- **Error type:** Length error (sequence too short by 408 tokens)

Titans MAL prediction: [**Correct**] *1,618 tokens (99.75% of target)*

- First 1,500 tokens: Near-perfect match
- Tokens 1,501–1,618: Minor coefficient errors (e.g., `16` instead of `8`) but correct structure
- All 47 Mandelstam variables present in correct positions
- **Result:** Token accuracy 98.2%; exact match [**Correct**]

Analysis: This example demonstrates the Titans MAL's ability to maintain long-range structural tracking. The memory module stores high-level constraints ("47 Mandelstam terms expected") that prevent premature termination.

D.3 Incorrect Predictions: Structural Errors

D.3.1 QED Example 2: Incorrect Parenthesis Nesting

Ground truth (172 tokens):

```
1/4*e^4*((s_12+m_e^2)*(s_13+m_mu^2)+(s_14+m_e^2)*(s_23+m_mu^2))
```

Titans MAL prediction (172 tokens): [Failed]

```
1/4*e^4*(s_12+m_e^2*(s_13+m_mu^2)+(s_14+m_e^2)*(s_23+m_mu^2))
```

Difference: Missing closing parenthesis after first `m_e^2`, creating incorrect algebraic grouping: $(s_{12} + m_e^2(s_{13} + \ldots))$ instead of $(s_{12} + m_e^2)(s_{13} + \ldots)$

Error type: Structural error (incorrect nesting)

Physics validity checks:

- Balanced parentheses: **[Pass]** (equal counts)
- Correct structure: **[Fail]** (wrong grouping changes mathematical meaning)
- Coupling constant: **[Pass]** (e^4 correct)

Analysis: Even with memory augmentation, rare patterns (specific parenthesis nesting) can be mislearned from limited training data (288 QED samples). This suggests data augmentation (e.g., swapping terms to generate variants) could improve robustness.

D.4 Incorrect Predictions: Coefficient Errors

D.4.1 QCD Example 3: Wrong Numerical Coefficient

Ground truth:

```
1/9*g^4*(16*m_u^2*m_d^2+s_12*s_13+[...])
```

Titans MAL prediction:

```
1/6*g^4*(16*m_u^2*m_d^2+s_12*s_13+[...])
```

Difference: Leading coefficient `1/9` predicted as `1/6`

Error type: Coefficient error (structure correct, constant wrong)

Analysis: Feynman rule combinatorics produce specific fractional prefactors ($1/9 = 1/3^2$ from color averaging). The model generates a plausible fraction (`1/6`) but incorrect for this process. This indicates incomplete learning of QCD color factor rules.

D.5 Summary of Error Patterns

Key takeaway: Titans MAL significantly reduces QCD structural errors ($3 \to 1$) while maintaining low length errors (2 remain). This validates that memory helps with algebraic structure tracking but does not fully eliminate positional encoding limitations.

E Reproduction Instructions

This appendix provides step-by-step instructions to reproduce all experiments reported in this work.

E.1 System Requirements

Hardware:

- **QED training:** Apple M3 Air (8GB RAM) with MPS acceleration, or any GPU with ≥4GB VRAM
- **QCD training:** NVIDIA T4/V100/A100 GPU with ≥16GB VRAM (Google Colab T4 sufficient)
- **Storage:** ∼500MB for dataset, preprocessed files, and model checkpoints

Software:

- Python 3.11 or later
- PyTorch 2.6.0 with CUDA 12.1 (or MPS for Apple Silicon)
- Additional packages: NumPy, pandas, scikit-learn, matplotlib, tqdm

E.2 Environment Setup

E.2.1 Using Conda (Recommended)

```
# Create environment
conda create -n symba-eval python=3.11
conda activate symba-eval

# Install PyTorch (adjust for your CUDA version)
conda install pytorch torchvision torchaudio pytorch-cuda=12.1 \
    -c pytorch -c nvidia

# Install dependencies
pip install -r requirements.txt
```

Listing 2: Conda environment setup

E.2.2 Using pip

```
python3.11 -m venv symba-env
source symba-env/bin/activate  # On Windows: symba-env\Scripts\activate
pip install torch torchvision torchaudio --index-url \
    https://download.pytorch.org/whl/cu121
pip install -r requirements.txt
```

Listing 3: pip environment setup

E.3 Repository Structure

```
evaluation-task/
|-- README.md                  # Project overview
|-- ARCHITECTURE.md            # Design rationale
|-- RESULTS.md                 # Complete results with CI
|-- requirements.txt           # Python dependencies
|-- dataset_analysis.py        # Tokenizer + analysis script
|-- notebooks/
```

```
|   |-- notebook1_preprocessing.ipynb          # Task 1.2
|   |-- notebook2_transformer_baseline.ipynb   # Task 1.3
|   +-- notebook3_titans_mal.ipynb             # Task 1.4
|-- data/
|   +-- processed/
|       |-- tokenizer.pkl                      # Custom tokenizer
|       |-- qed_train.pkl, qed_val.pkl, qed_test.pkl
|       |-- qcd_train.pkl, qcd_val.pkl, qcd_test.pkl
|       +-- data_profile.json                  # Dataset statistics
+-- weights/
    |-- baseline_qed_best_ckpt.pt
    |-- baseline_qcd_best_ckpt.pt
    |-- titans_mal_qed_best_ckpt.pt
    |-- titans_mal_qcd_best_ckpt.pt
    |-- baseline_qed_training_log.csv
    |-- baseline_qcd_training_log.csv
    |-- titans_qed_training_log.csv
    +-- titans_qcd_training_log.csv
```

E.4 Step-by-Step Reproduction

E.4.1 Step 1: Data Preprocessing (Task 1.2)

Run `notebook1_preprocessing.ipynb` or execute:

```
python dataset_analysis.py
```

Outputs:

- `data/processed/tokenizer.pkl`: Custom tokenizer (192 tokens)
- `data/processed/qed_*.pkl`: QED train/val/test splits (288/36/36)
- `data/processed/qcd_*.pkl`: QCD train/val/test splits (187/23/24)
- `data/processed/data_profile.json`: Dataset statistics

Verification:

```
import pickle
with open('data/processed/tokenizer.pkl', 'rb') as f:
    tokenizer = pickle.load(f)
print(f"Vocabulary size: {tokenizer.vocab_size}")  # Should print 192
```

E.4.2 Step 2: Baseline Transformer Training (Task 1.3)

QED Baseline:

```
# Open notebook2_transformer_baseline.ipynb
# Set: PHYSICS_MODEL = 'QED', MAX_EPOCHS = 150
# Run all cells
```

QCD Baseline:

```
# Open notebook2_transformer_baseline.ipynb
# Set: PHYSICS_MODEL = 'QCD', MAX_EPOCHS = 750
# Set: BATCH_SIZE = 4, GRAD_ACCUM_STEPS = 4
# Run all cells
```

Expected outputs:

- QED: 88.89% test EM, 97.41% token accuracy
- QCD: 79.17% test EM, 81.67% token accuracy

Training time:

- QED: ∼2 hours (Apple M3 MPS)
- QCD: ∼18 hours (Colab T4 GPU)

E.4.3 Step 3: Titans MAL Training (Task 1.4)

QED Titans MAL:

```
# Open notebook3_titans_mal.ipynb
# Set: PHYSICS_MODEL = 'QED', MAX_EPOCHS = 150
# Run all cells (memory module code included in notebook)
```

QCD Titans MAL:

```
# Open notebook3_titans_mal.ipynb
#Set: PHYSICS_MODEL = 'QCD', MAX_EPOCHS = 750
# Set: BATCH_SIZE = 4, GRAD_ACCUM_STEPS = 4
# Run all cells
```

Expected outputs:

- QED: 83.33% test EM, 97.34% token accuracy
- QCD: 83.33% test EM, 97.06% token accuracy

Training time:

- QED: ∼2.5 hours (Apple M3 MPS)
- QCD: ∼21 hours (Colab T4 GPU)

E.5 Reproducing Specific Results

E.5.1 Bootstrap Confidence Intervals

From any notebook after evaluation:

```python
import numpy as np
from sklearn.utils import resample

# Bootstrap 95% CI for exact match accuracy
def bootstrap_ci(correct_mask, n_resamples=10000, alpha=0.05):
    accuracies = []
    for _ in range(n_resamples):
        sample = resample(correct_mask)
        accuracies.append(sample.mean())
    lower = np.percentile(accuracies, 100 * alpha / 2)
    upper = np.percentile(accuracies, 100 * (1 - alpha / 2))
    return lower, upper

# Usage:
correct = np.array([True, True, False, True, ...])  # Test results
ci = bootstrap_ci(correct)
print(f"95% CI: [{ci[0]:.1%}, {ci[1]:.1%}]")
```

E.5.2 Error Taxonomy Analysis

```
def categorize_error(pred_tokens, target_tokens):
    if len(pred_tokens) < 0.9 * len(target_tokens) or \
       len(pred_tokens) > 1.1 * len(target_tokens):
        return 'length'
    # Check structure by comparing operators/parentheses
    pred_struct = [t for t in pred_tokens if t in {'(', ')', '+', '*'}]
    tgt_struct = [t for t in target_tokens if t in {'(', ')', '+', '*'
        }]
    if pred_struct != tgt_struct:
        return 'structural'
    return 'coefficient'

# Apply to test set
for pred, target in zip(predictions, targets):
    if pred != target:
        error_type = categorize_error(pred, target)
        print(f"Error type: {error_type}")
```

E.6 Pre-trained Model Checkpoints

All trained model checkpoints are available in `weights/` directory:

- `baseline_qed_best_ckpt.pt`: Baseline QED (88.89% EM)
- `baseline_qcd_best_ckpt.pt`: Baseline QCD (79.17% EM)
- `titans_mal_qed_best_ckpt.pt`: Titans MAL QED (83.33% EM)
- `titans_mal_qcd_best_ckpt.pt`: Titans MAL QCD (83.33% EM)

Loading a checkpoint:

```
import torch
model = TransformerModel(...)  # Initialize with correct architecture
checkpoint = torch.load('weights/titans_mal_qcd_best_ckpt.pt')
model.load_state_dict(checkpoint['model_state_dict'])
model.eval()
```

E.7 Troubleshooting

Out of memory on QCD training:

- Reduce batch size to 2 with `GRAD_ACCUM_STEPS=8`
- Use `torch.cuda.empty_cache()` between epochs
- Enable gradient checkpointing (see notebook comments)

Reproducibility issues (different random results):

- Ensure all random seeds are set:

```
torch.manual_seed(2024)
np.random.seed(42)
torch.use_deterministic_algorithms(True)
```

- Note: Some operations (e.g., `torch.scatter_add_`) may not have deterministic implementations

Tokenizer UNK tokens:

- Verify `max_idx=120` in tokenizer initialization
- Check that single digits `0`, `5`, `6`, `7`, `9` are in vocabulary

E.8 Additional Resources

- **GitHub Repository:** `https://github.com/evenindividual04/gsoc2026-symba-titans`
- **Dataset:** SYMBA data files in `SYMBA - Test Data/common-task-1.2/`
- **Questions/Issues:** Open an issue on GitHub or contact `f20221215@goa.bits-pilani.ac.in`

Table 17: Complete baseline Transformer hyperparameters

Parameter	**QED**	**QCD**
Architecture		
Vocabulary size	192	192
Embedding dimension (d_{model})	128	128
Encoder layers	3	3
Decoder layers	3	3
Attention heads (n_{head})	4	4
Head dimension	32	32
FFN dimension	512	512
Dropout	0.1	0.1
Activation	ReLU	ReLU
Layer norm type	Pre-LN	Pre-LN
Positional encoding	Sinusoidal	Sinusoidal
Max position	2200	2200
Training		
Optimizer	AdamW	AdamW
Learning rate	3×10^{-4}	3×10^{-4}
Weight decay	1×10^{-4}	1×10^{-4}
β_1	0.9	0.9
β_2	0.999	0.999
ϵ	1×10^{-8}	1×10^{-8}
Batch size	16	4
Gradient accumulation	1	4
Effective batch size	16	16
Max epochs	150	750
Warmup epochs	15	75
Min LR	1×10^{-6}	1×10^{-6}
LR schedule	Linear + cosine	Linear + cosine
Gradient clipping	1.0	1.0
Label smoothing	0.0	0.0
Early stopping patience	25	25
Decoding		
Strategy	Greedy	Greedy
Max decode length	1700	1700
Total parameters	**1,437,696**	**1,437,696**

Table 18: Complete Titans MAL hyperparameters (incremental to baseline)

Parameter	**QED**	**QCD**
(All baseline parameters as above, plus:)		
Memory Module		
Memory dimension (d_{mem})	64	64
Memory learning rate (η_{mem})	5×10^{-3}	5×10^{-3}
Memory momentum (β)	0.7	0.7
Memory activation	GELU	GELU
Memory initialization	Xavier uniform	Xavier uniform
Gate α init	0.0	0.0
Memory reset	Per-sequence	Per-sequence
Surprise target	detach($\mathbf{h}$)	detach($\mathbf{h}$)
Update rule	Gradient descent	Gradient descent
Memory parameters	**+149,763**	**+149,763**
Total parameters	**1,587,459**	**1,587,459**
Parameter overhead	**+10.4%**	**+10.4%**

Table 19: Token usage statistics across full SYMBA dataset

Token Category	**Unique Tokens**	**Total Occurrences**
Algebraic operators	11	98,247
Index tokens (IDX_*)	120	87,532
Numbers	24	45,681
Mandelstam variables	6	23,145
Masses	8	12,389
Coupling constants	2	8,764
Physics operators	3	7,234
Particle symbols	13	6,912
Momenta	4	4,270
Special tokens	4	0 (not in data)
Total	**192**	**294,174**

Table 20: Error pattern frequency across test sets

Error Type	**Baseline QED**	**Baseline QCD**	**Titans QED**	**Titans QCD**
Length (sequence too short/long)	2 / 4	2 / 5	2 / 6	2 / 4
Structural (wrong nesting/operations)	2 / 4	3 / 5	3 / 6	1 / 4
Coefficient (wrong constants)	0 / 4	0 / 5	1 / 6	1 / 4